CHAGEE

CHAGEE SINGAPORE

FORECAST THE PEAK, SHARE THE COUNTER

PREDICTIVE AND PHYSICAL AI: WHERE TO OPEN NEXT, AND WHAT THE GM DECIDES FOR THE BUSIEST HOUR

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



SPEAKER NOTES · TITLE

Good morning. We are Group 3. Our company is Chagee, and we look at one place in its business: the counter of a Singapore outlet in its busiest hour.

Two kinds of technology meet there. Predictive AI forecasts demand: where to open, and how busy the hour will be. Physical AI, automation and machines, does the fixed steps. Around them, people share the hand steps like a bucket brigade.

Our decision-maker throughout is the GM, Chagee Singapore's general manager. Timing: title and agenda together, about twenty seconds.

Agenda

01 **Chagee in Singapore**
How it earns and sells

02 **From farm to cup**
Where the supply chain is stable, and where it is not

03 **The data trail**
What every cup already records

04 **Our scope**
Two areas, two kinds of AI, one decision-maker · 04b network effects: what to grow, what to cap

05 **Predict demand**
App searches, DBSCAN and a new-store pin

06 **Share the counter**
The bucket brigade, side by side

07 **Split the counter**
Which steps stay human: the GM's call

08 **Four scores for every location** Extension
A pin rated on four things · 08b the supervised model, as a proxy for a new market

09 **From pin to net profit** Extension
Six steps to a go / no-go, net to Chagee

10 **The GM's playbook** TBC
Four decisions and a pilot

11 **The size of the benefit** TBC
Orders kept, revenue, and what it rests on



SPEAKER NOTES · AGENDA

Eight short chapters. The first four set the scope; the last four are the forecast, the counter, the GM's split and the playbook. Click any chapter to jump there.

In Singapore, outlets run under 1 company, and every channel ends at 1 counter

One company, one data system: order and app data sit together

HOW A SINGAPORE CUP IS SOLD

GROUP NET REVENUE, FY 2025



Franchise **88%** Company-owned **12%**, growing fastest

SG Singapore outlets run under one company and one data system (Chagee source, anonymous)

In store	Walk-in (app order; tourists can order at the counter, to confirm)
Chagee app	order ahead, collect
Delivery	third-party delivery apps (e.g. Grab); no Chagee-run delivery

One counter every channel's cup is made here



44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **One company, one data system**

  Singapore outlets run under **1 company** and a **single data system**.

  Opportunity: A counter design or roster rule rolled out across outlets.

  Chagee source, anonymous

- **Chagee-App**

  An **order-ahead, self-pickup app**; in Singapore, **in-app ordering and delivery** followed the reopening queues. Chagee **collects personal and transactional data**.

  To confirm: the app's share of Singapore orders; whether searches or location are logged.

  · CHAGEE App Store listing · Vulcan Post, Oct 2024 · Chagee Form 20-F, FY 2025

- **Why Chagee?**

  A **fast-growing, playful** brand with **heavy app use**.

  Likely to have **rich order data** and **room for AI** to help the stores.

TAKEAWAY | Whatever the channel, every order ends at one counter. Peak hour is where an outlet wins or loses.

SPEAKER NOTES · CHAGEE IN SINGAPORE

Step through four points. One: at group level most revenue comes from franchising, about 88% in FY 2025; company-owned is about 12% and growing fastest. That split is from a secondary source.

Two: Singapore outlets run under one company and one data system. We have this from a Chagee source who asked not to be named; please do not identify them. They told us the ownership form does not matter here because the data sits in one system. That a counter design or roster rule could be rolled out across outlets is our reading, not something we were told.

On the app card: Chagee has an order-ahead, self-pickup app, per its App Store listing, and Vulcan Post reports that Chagee Singapore rolled out in-app ordering and delivery after the long reopening queues. Its Form 20-F for FY 2025 says the business generates and processes personal and transactional data. What we cannot back publicly: the app's share of Singapore orders, and whether searches or location are logged; the App Store privacy label lists precise location as not linked to the user. Those are to confirm with Chagee.

Three: 44 outlets, by the store directory in October (press reports vary: 44–46). Chagee prefers ground-floor mall units.

Four: walk-in, app orders for collection, and third-party delivery orders all end at the same counter. Ordering is app-first: walk-ins order in the app, and the source mentioned counter ordering only as an exception for tourists, which we will confirm. The Chagee app is for ordering ahead and collecting; all delivery goes through third-party apps such as Grab, by the same Chagee source, and there is no Chagee-run delivery. That is why we look at the counter. About fifty seconds.

Supply is standard and pre-planned. Counter feels demand within minutes

Tea leaves travel one standard route. Only fresh milk and fruit are bought locally.

FOLLOW ONE CUP · STEP THROUGH



Source

Tea leaves from **Yunnan** through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

Outlet Counter

Machines: brew each cup from a scanned recipe
People: Ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times (to confirm)

Last Mile

Self-collect via app, or delivery via **third-party** apps such as Grab.
Riders wait at the same counter

DATA CREATED

Orders by channel

Rider wait (to confirm)

Customer

Orders through app and **collects** at the counter.

DATA CREATED

App orders, drop-offs

- **Stable upstream**
  Standard sourcing, regional warehouses and forecast-led restocking: Chagee already plans this part.
  YF Chuhai; Chagee Form 20-F
- **Volatile downstream**
  Demand at a counter **moves by the hour**: lunch, a mall event, rain, a new drink launch.
- **So we stop at the counter**
  Question is: How many people the counter needs this hour? And how do they work? (Not shipment)

TAKEAWAY | Chagee already plans its stock (public sources). People are deployed on local expertise, by the GM and outlet teams (Chagee source, anonymous); that is the call we model.

SPEAKER NOTES · FROM FARM TO CUP

Press play and the cup travels farm to customer. Sourcing is standard; only fresh milk and fruit are local (YF Chuhai). Chagee runs central and regional warehouses (Form 20-F) and already plans restocking from forecasts.

The volatile part is the counter, the red chips. At the counter, a Chagee source who asked not to be named told us the GM and outlet teams deploy people on local expertise; we model how a forecast could support those calls. Under time pressure, one sentence: supply is planned ahead; the counter runs on local judgement. About forty seconds.

Orders are already recorded; the other inputs a forecast needs are candidates to confirm

Four data sources and six decisions they could feed. We follow the four that start in the app and end at the counter.

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

	Store and production	App and members	Delivery	Supply and context
Demand forecasts	Cups per hour by drink and channel	Member repeat demand	Orders per hour by platform	Weather, holidays, launches
Staffing and prep	Wait times, machine logs	Pickup no-shows	Hand-over peaks	Labour hours vs cups per hour
Replenishment	Drink mix per outlet	Launch trial rates	Group-order toppings	Ingredient use, expiry, par
Queue and capacity	Wait time, footfall vs sales	Store-select drop-offs	Rider wait at counter	Stock-outs at peak
Site selection	Sales density per outlet	"No store nearby", home and work	Distance and fee by postcode	Mall footfall, competitors
Menu and promos	Drink share, customisation	Trial and repeat, voucher uplift	Delivery basket mix	Collab calendar

White chips: core input; grey: refines it; red rows: our scope. Wait times, footfall, searches and location are to confirm.

- **Known and to confirm**

  Known: orders by channel and time (point of sale, app, Grab).

  To confirm: searches, customer location, wait times and walk-aways; some may need new logging.

- **The app is the link**

  Only the app ties a cup to a person. That needs consent and care under Singapore's data protection law (PDPA).

- **What we take forward**

  Demand forecast, then staffing, then queue capacity: one chain that ends with the GM's counter and roster.

TAKEAWAY | Orders by hour are recorded; searches, location, waits and walk-aways need confirming. Today, GM and outlet teams deploy people on local expertise (Chagee source, anonymous); we show how a forecast could support those calls.

SPEAKER NOTES · THE DATA TRAIL

Each step lights up one decision and the data that feeds it. Skip to the last step under time pressure: our scope is the three counter rows: forecast, staffing, queue.

This list is our reading of what Chagee's systems could record. Orders by channel and time are known; searches, customer location, wait times and walk-aways are candidates to confirm, based on public material and our industry conversation, not a Chagee data dictionary. About fifty seconds.

New stores and the in-store peak, for one decision-maker: the GM

Two of our three ideation areas, enabled by two kinds of AI, each simulated with the trade-off the GM weighs.

FROM THREE IDEAS TO TWO · STEP THROUGH

NOT COVERED YET

Online retailing

Steer orders between app pickup and Grab by hour.

To be added. Needs platform fees and price tests we do not have.

OUR FOCUS · EXPANSION

New stores

Map where the next outlet adds the most cups, from app searches.

Rent, fit-out and payback stay future work.

Topic 3 · forecasting

OUR FOCUS · OPERATIONS

In-store operations

Forecast the peak, set the roster, share the counter.

Order data exists; the decision is daily.

Topic 3 · forecasting

Topic 5 · bucket brigades

AI · PREDICTIVE

Forecast demand: where a new outlet adds cups, and cups per hour for the roster.

AI · PHYSICAL

Automation and machines at the counter, from dispensers up to machines that make the drink.

OPERATIONS TOOLS, NOT AI

Bucket brigade and cross-training: how people share the hand steps.

The GM

General manager, Chagee Singapore

DECIDES	Where to open next, which counter steps go to machines, how many people work the peak, and how they share it
WHEN	Before a store opens, and when rosters are set
USES	The forecasts and the simulations
TODAY	People deployed on local expertise: the GM and outlet teams make the calls (Chagee source, anonymous)

SIMULATED, WITH THE TRADE-OFF THE GM WEIGHS

PREDICT	cups added vs cups moved from nearby outlets
SHARE	bucket brigade vs fixed posts
SPLIT · MACHINES	shorter wait vs higher cost
BENEFIT	orders kept vs extra labour

TAKEAWAY | Predictive AI says where to open and how many people the peak needs; physical AI changes which steps machines do; the bucket brigade decides who does the rest.

SPEAKER NOTES · OUR SCOPE

Our ideation produced three areas. Online retailing is not covered yet and will be added; for now we take two: new stores, for expansion, and the in-store peak.

AI comes in two kinds here. Predictive AI forecasts demand: where a new outlet adds cups, and how busy each hour will be. Physical AI is automation and machines at the counter, from dispensers up to machines that make the drink; that is a type of machine, not a claim about Chagee's plans. Bucket brigade and cross-training are operations tools from Topic 5, order fulfilment, but they decide whether a lean roster actually works.

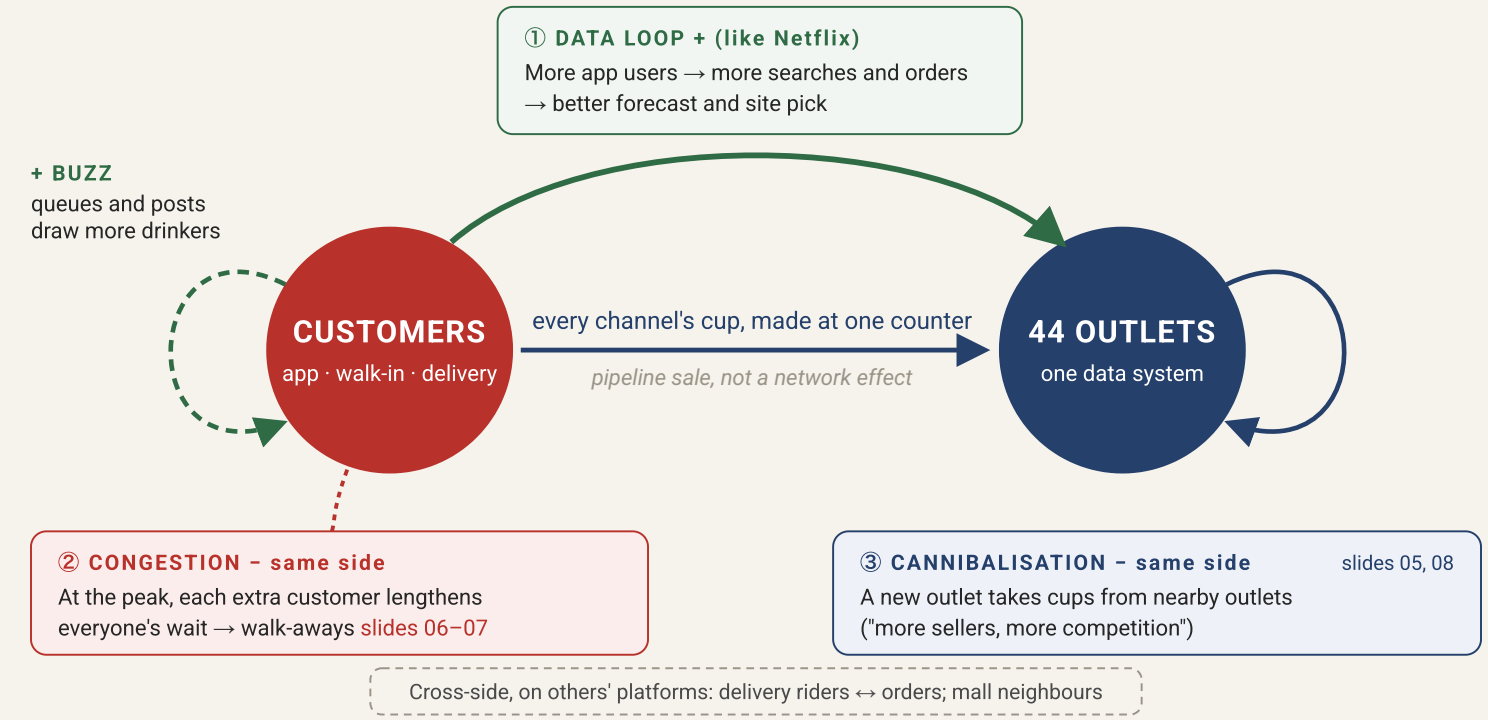
Every part of the deck is a simulation with a trade-off the GM has to weigh: cups added against cups moved, brigade against fixed posts, a shorter wait against a higher cost, and orders kept against extra labour. The decision-maker is the GM. Today the GM and outlet teams deploy people on local expertise, according to an anonymous Chagee source; the forecast is meant to support those calls, not replace them. About fifty seconds.

Course link · cross-side vs same-side network effects (eBay)

Chagee is not a two-sided platform; its network effects sit on one side, and our AI works on three of them

Our reading: Chagee sells its own tea, so there is no seller side to match. One data loop to grow, two crowding effects to cap.

WHERE CHAGEE SITS ON THE DIAGRAM · CLICK A LOOP



① GROW · PREDICTIVE AI

Feed the data loop

Why Chagee: app-first and member-led. Group-wide, **257 million** registered members (end of June 2026), and members with two or more purchases place **more than 78%** of orders (earnings call, Aug 2026; SG share to confirm).

Site pick: **+731** vs **+216** cups a day (synthetic).

② CAP · FORECAST + COUNTER

Absorb the peak crowd

Forecast crew plus a bucket brigade: event day walk-aways **54%** → **1.2%** of peak-hour orders.

③ CAP · SITE SCORING

Count cups added, not sold

Of 837 cups at the new site, **106** (13%) move from nearby outlets. The score subtracts them.

NOT OUR LEVER

Delivery apps and malls own the cross-side effects; their riders still count in our peak.

Takeaway | More customers help Chagee only if the map and the counter can absorb them: predictive AI grows the data loop; the forecast, counter design and site score cap congestion and cannibalisation.

SPEAKER NOTES · NETWORK EFFECTS

These two areas are connected: more customers generate more data, but they also create queues, while new outlets can take sales from existing ones. Our models examine those trade-offs. A lens from the course: on eBay, buyers and sellers attract each other: a cross-side network effect. Chagee is not that, in our reading: it sells its own tea, so there is no seller side to match. Customers to outlets is a pipeline sale.

What it does have sits on one side. One effect to grow: the data loop, like Netflix. More app users give more searches and orders, so the forecast and the site pick get sharper. This is why we chose Chagee: its business model is app-first and member-led. On its second-quarter 2026 earnings call, Chagee said it had 257 million registered members at the end of June, and that members who made two or more purchases accounted for more than 78% of total orders. Those members sit on Chagee's mobile mini program, its ordering app, according to its first-quarter 2025 results. So nearly every cup leaves a data trail that can feed the loop. These are group-wide figures, not Singapore; Singapore's app share of orders is still to confirm. A same-side plus, buzz: queues and posts draw more drinkers; we do not model it.

And two crowding effects to cap. Congestion: at the peak, every extra customer lengthens everyone's wait; with a normal-day roster on an event day, about 54% of peak-hour orders walk away, and the forecast crew with a bucket brigade brings it to about 1.2%. Cannibalisation: a new outlet takes cups from its neighbours; of 837 cups at the app-search site, 106 move from nearby outlets, against far more beside an existing outlet.

Cross-side effects exist, but on others' platforms: delivery apps and malls. Not our lever. About forty-five seconds.

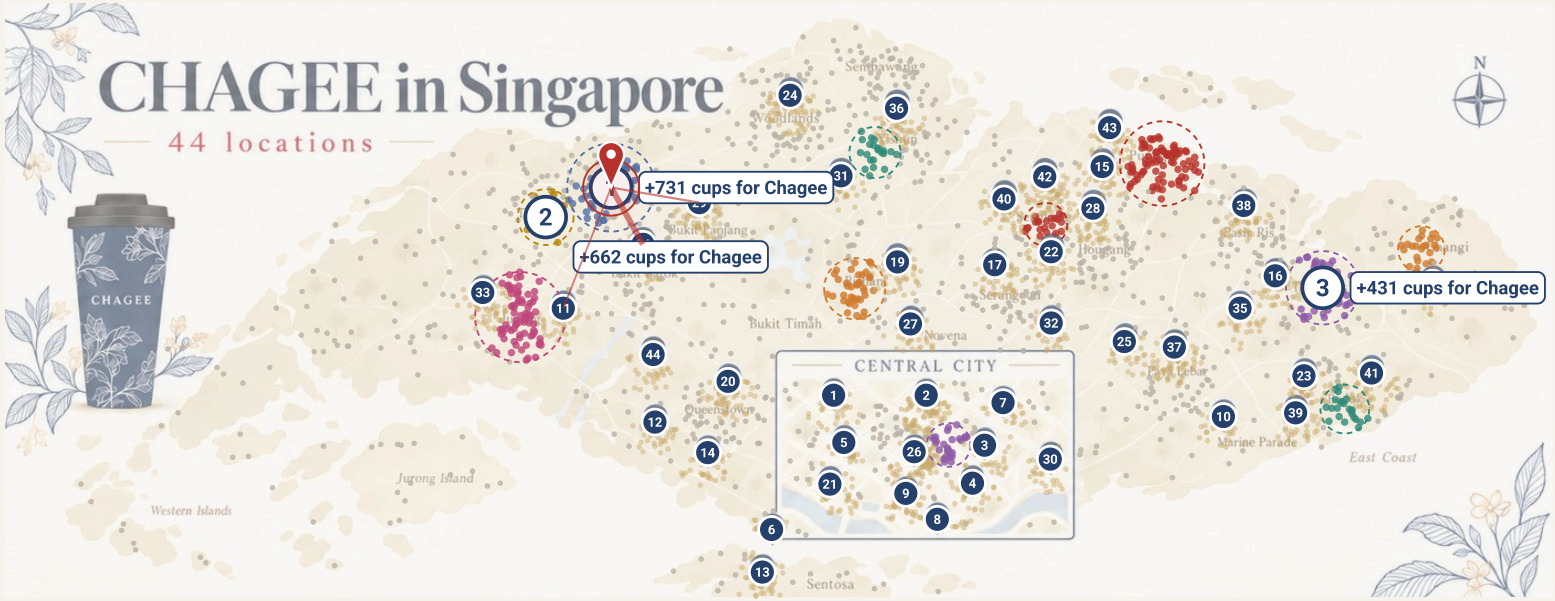
Course link · Week 3, Analytics and AI · the Flashion new-store forecast

Where would Outlet #45 add the most cups for Chagee?

Not what a new store sells, but what it adds once you subtract the cups it takes from nearby outlets.

CONTEXT Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 06–07).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method.

WITHOUT AI our assumed baseline

Open outlet 45 beside 34 · West Mall: it sells **604** cups a day, but 64% are moved from outlets nearby, so **+216** for Chagee.

WITH AI our proposal

DBSCAN clusters synthetic app searches; the assumed demand model scores the same outlet 2.7 km away, near Choa Chu Kang: sells 837, **+731** for Chagee.

SPEAKER NOTES · PREDICT DEMAND FOR A NEW STORE

Chagee Singapore runs on its app, with few kiosks. If Chagee logs searches and their location, which we have to confirm, every search is a demand signal, including searches where no store is close. Here the searches are synthetic. That makes the app the forecasting input for a new store, before it has any sales.

Step one: synthetic app sessions on our map. Gold are within walking distance of an outlet; red found no store nearby. Step two: DBSCAN, unsupervised machine learning, groups red searches that are dense; scattered ones are noise. Step three: cluster centres are candidate sites.

Step four: click the map to test a site. The card leads with the gain for Chagee, which is the store's sales minus the cups it takes from nearby outlets, and gives a verdict: strong, mixed, or mostly moving cups. Try a pin next to Tampines West CC: it sells, but most of it moves from the two Tampines outlets nearby. Step five: the forecast's peak hour gives the opening roster for the counter on the next slides.

If asked how DBSCAN works, press How it works: it replays the algorithm one search at a time on a toy set. The appendix lists every simulated number.

The three sliders are the only inputs: cluster radius, minimum searches per cluster, and how far people will walk. The data is synthetic; with Chagee's real app logs, the same steps would run unchanged. About one and a half minutes.

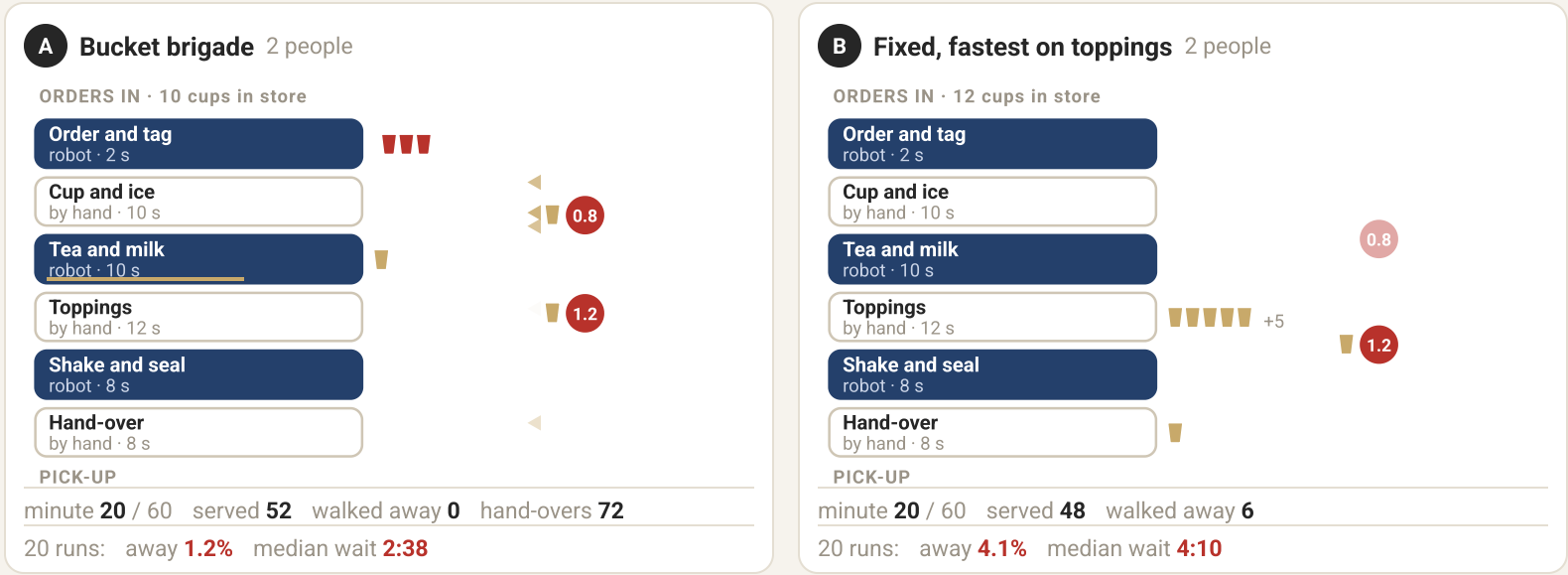
Course link, Week 3, Analytics and AI: in the Flashion example, demand for a new store was forecast from similar stores. We ask the same question with a different input: app searches around the site.

Course link · Week 5 and Week 6, equipment and technology · the bucket-brigade simulation

With a lean roster, a bucket brigade keeps cups moving without knowing who is fastest

Today's split (our scenario), two ways of sharing the hand steps, side by side. Pick what each side runs.

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES



Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

ALL THREE RULES, MEAN OF 20 RUNS

EVENT DAY, 2 PEOPLE	WALKED AWAY	MEDIAN WAIT
Bucket brigade	**1.2%**	**2:38**
Fixed, fastest on toppings	4.1%	4:10
Fixed, slowest on toppings	34.9%	9:47

At the people on side A. A customer leaves if the wait would pass 5 min. With spare people, a well-assigned fixed line is as good or a little faster.

Why lean is where it counts

Work flows to whoever is free, so the line balances itself (Topic 5). Fixed stations are only as good as the guess of who is fast.

Machines = automation, each side's equipment, held fixed. **AI = AI-assisted staffing**: the forecast sets the crew; customers and sharing rule unchanged. Event day · peak hour · mean of 20 runs · illustrative assumptions

WITHOUT AI INTELLIGENCE normal-day roster	**A** Bucket brigade: 1 person · away **53.8%** · wait 7:24 · S$372/day **B** Fixed, fastest on toppings: 1 person · away **54.2%** · wait 7:28 · S$372/day
WITH AI INTELLIGENCE forecast-informed roster	**A** 2 people · away **1.2%** · wait 2:38 · S$402/day · +1 staff, −52.6 pts away, +S$30 **B** 2 people · away **4.1%** · wait 4:10 · S$402/day · +1 staff, −50.1 pts away, +S$30

SPEAKER NOTES · SHARE THE COUNTER

Same counter, same customers, two sharing rules side by side. The counter is today's split, our scenario, not Chagee's disclosed setup. Robots, the navy boxes, take the order, brew tea and milk, and shake and seal. People do cup and ice, toppings and hand-over.

Left, a bucket brigade: each person carries a cup forward; the last hands it over, walks back and takes over a colleague's cup. Right, fixed stations: each person owns a station.

Press play on an event day. Then set the right side to "slowest on toppings": the queue builds at toppings and customers walk away. The brigade never needed to know who was fast.

Be honest about the other case: add people and fixed stations catch up, or are a little faster. About one minute.

Then click the band to reveal the AI layer on its own. Machines are automation, and here they stay exactly as they are: today's split on both sides. AI means AI-assisted staffing: the demand forecast sets the crew. Equipment, customers and each side's sharing rule are held fixed; only the crew changes. Without AI, the normal-day roster of 1 faces an event day: about 54% of peak-hour orders walk away with a brigade, 54% with fixed posts. With AI, the forecast crew of 2 brings that to 1.2% and 4.1%, for S$402 a day against S$372. The band follows the day you pick. Walk-aways and wait are the peak hour, mean of 20 runs; cost is the day's roster plus machines and leaves out lost sales. All illustrative assumptions.

Course link: the bucket-brigade simulation from Week 5 and Week 6, equipment and technology, inspired this counter replica. Same idea, a different use case.

Course link · Week 5 and Week 6, equipment and technology

In our model, one tea-and-milk machine is the cheapest on every day type Pilot this against today's split.

PEAK HOUR · SAME CUSTOMERS

A All by hand 4 people, brigade

ORDERS IN · 0 cups in store

- Order and tag — by hand · 5 s
- Cup and ice — by hand · 10 s
- Tea and milk — by hand · 15 s
- Toppings — by hand · 12 s
- Shake and seal — by hand · 10 s
- Hand-over — by hand · 8 s

0.8 0.9 1.1 1.2

PICK-UP

minute **20** / 60 served **57** walked away **0** hand-overs **156**

20 runs: away **0.0%** wait **1:25** **S$422**/day

B Machine-made drink (scenario) 2 people, fixed posts

ORDERS IN · 0 cups in store

- Order and tag — by hand · 5 s
- Cup and ice — robot · 8 s
- Tea and milk — robot · 10 s
- Toppings — robot · 15 s
- Shake and seal — robot · 8 s
- Hand-over — by hand · 8 s

0.8

1.2

PICK-UP

minute **20** / 60 served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

FIVE OPTIONS, MEAN OF 20 RUNS

EVENT DAY	PEOPLE	WALKED AWAY	WAIT	COST/DAY
Today's split (our scenario)	2	1.2%	2:38	S$402
Machine-made drink (scenario)	2 fixed	0.0%	1:06	S$465
All by hand	4	0.0%	1:25	S$422
All robots	–	0.0%	0:58	S$562
One machine (tea and milk) ✓	**3**	**0.2%**	**1:44**	**S$335**

✓ cheapest within the limit · fastest: All robots

Normal day, cost a day: One machine (tea and milk) S$305 (cheapest) · All by hand S$361 · Today's split (our scenario) S$372 · Machine-made drink (scenario) S$435 · All robots S$562.

Machines = automation, each side's equipment, held fixed. **AI = AI-assisted staffing**: the forecast sets the crew; customers and sharing rule unchanged. Event day · peak hour · mean of 20 runs · illustrative assumptions

WITHOUT AI INTELLIGENCE — normal-day roster

A All by hand: 2 people · away **40.1%** · wait 5:33 · S$361/day
B Machine-made drink (scenario): 1 person · away **7.2%** · wait 6:26 · S$435/day

WITH AI INTELLIGENCE — forecast-informed roster

A 4 people · away **0.0%** · wait 1:25 · S$422/day · +2 staff, −40.1 pts away, +S$60
B 1 person · away **7.2%** · wait 6:26 · S$435/day · the forecast adds no one here

GM DECISION Pilot one tea-and-milk machine against today's split (our scenario): in our model S$305–335 a day against S$372–402. That holds only while a machine unit costs more than about S$905 a month (same rate for its 1 unit and the split's 3; wage, volume and crews held fixed). We assume S$1,909; Chagee's real cost is unknown.

SPEAKER NOTES · SPLIT THE COUNTER

This is the GM's decision: which counter steps go to machines. Side A opens on human-only: people do every step. Side B opens on machine-made drink: machines make the drink, people tag the order and hand it over. Both are our illustrative scenarios, not Chagee's disclosed setup. Chagee's own equipment cost is not public; its Singapore relaunch used semi-automated equipment, according to Vulcan Post.

Keep two things apart. Machine automation is the split, the equipment on each side. AI here is AI-assisted staffing: the demand forecast sets the crew. Step four reveals the AI layer on its own. Each side keeps its equipment, customers and sharing rule; only the crew moves from the normal-day roster to the forecast roster. On an event day, all by hand without AI has 2 people, and about 40% of peak-hour orders walk away; with AI the forecast puts on 4, walk-aways fall to 0.0%, median wait 5:32 to 1:24, for S$422 a day against S$361. On the machine-made drink the forecast keeps 1 person, so AI changes nothing there: 7.2% still walk away, and the split table adds a person to meet the limit. In our model, forecasting matters most where people do the work. The band follows the day and the equipment you pick; walk-aways and wait are the peak hour, mean of 20 runs; cost is the day's roster plus machines, without lost sales. Illustrative assumptions throughout.

How to use it: choose a split for each side, or click any station to hand it to a robot or a person, then run the peak hour or press End to compare. Each side starts with the fewest people, from the forecast roster up, that keep walk-aways within 2%.

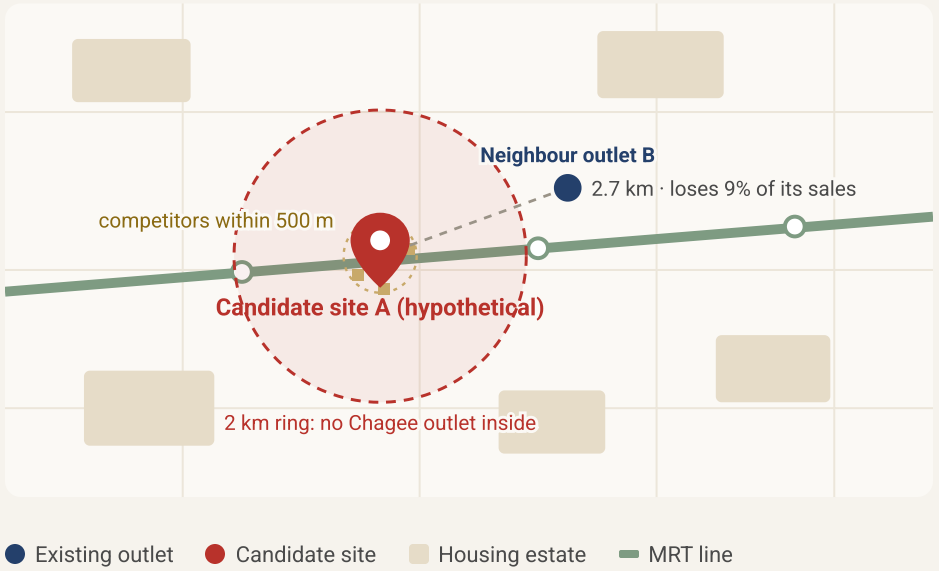
The table compares five options on the chosen day, mean of 20 runs: today's split, the machine-made drink, all by hand, all robots, and one tea-and-milk machine from the machine ladder in the appendix. In our model, one tea-and-milk machine is the cheapest of the five on every day type, because it lets the roster drop a person off-peak at the cost of one machine unit. All robots is the fastest and the dearest: among these options robots buy a shorter wait, not a lower cost.

Extension: where this scales next. Not included in the S$ benefit.

Four scores for every location

Corporate drops a pin; the model rates it on four things (illustrative). Decision-maker here: corporate expansion planning, not the GM.

SCHEMATIC MAP · HYPOTHETICAL PINS



Demand score HAND-SET
Residents, workers and transit riders within reach, plus footfall. — 84

Delivery demand score HAND-SET
Orders from the area on third-party delivery apps.
All Chagee delivery goes through third-party apps (anonymous Chagee source, to confirm), so we use area demand, not "served from a farther outlet". Whether the platforms share this data is to confirm. — 78

Competition score HAND-SET
Bubble-tea and cafe outlets within 500 m; dense competition lowers the score, a few prove demand. — 61

Cannibalisation factor COMPUTED · SLIDE 05 ENGINE
Site A: 13% of its cups would move from existing Chagee outlets; neighbour outlet B loses 9%. Deducted so the result is net to Chagee. — −13

Site score WEIGHTED
Demand 50%, delivery 20%, competition 30% (hand-set weights). — 76

ATTRIBUTES THE MODEL TAGS TO EACH PIN residents and workers within 1 km · MRT and bus tap-ins · footfall · delivery orders from the area · competing outlets within 500 m · Chagee outlets within 2 km · rent per sq ft · staff needed at peak

WHAT IS AI HERE Scoring a pin from its attributes would be a supervised model trained on outlets' results: a production idea. Here the three scores are set by hand, and the cannibalisation factor is computed by the slide 05 engine on synthetic data. Slide 08b runs the supervised model.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

SPEAKER NOTES · FOUR SCORES FOR EVERY LOCATION

An extension: where this scales next, and not part of the benefit figure. The decision-maker here is corporate expansion planning, not the GM. This builds on Jin's site-selection framework.

Step through it. Corporate drops a pin; the model tags its attributes; three scores follow. Demand, delivery demand and competition are set by hand here, and you can drag a bar to test another value: in production, a supervised model would score them. We call the second one delivery demand, not delivery relief, because all Chagee delivery runs through third-party apps, according to an anonymous Chagee source; whether the platforms share this data is to confirm.

The cannibalisation factor is computed by the same engine as slide 05: about 13% of site A's cups would move from existing outlets. Switch to the pin with an outlet inside the ring, or the one beside an outlet, and watch it jump. All synthetic. About thirty-five seconds.

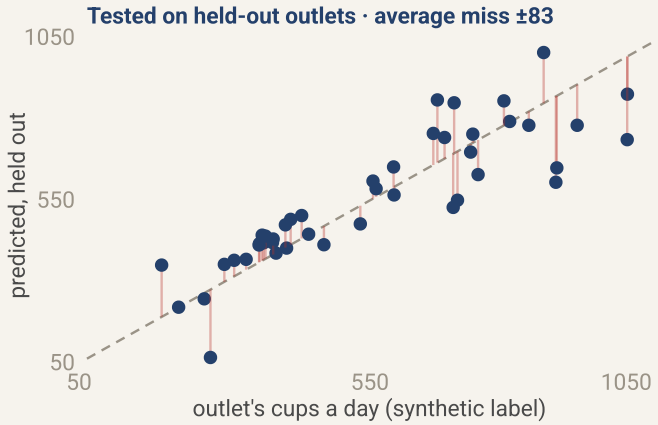
Extension: where this scales next. Not included in the S$ benefit.

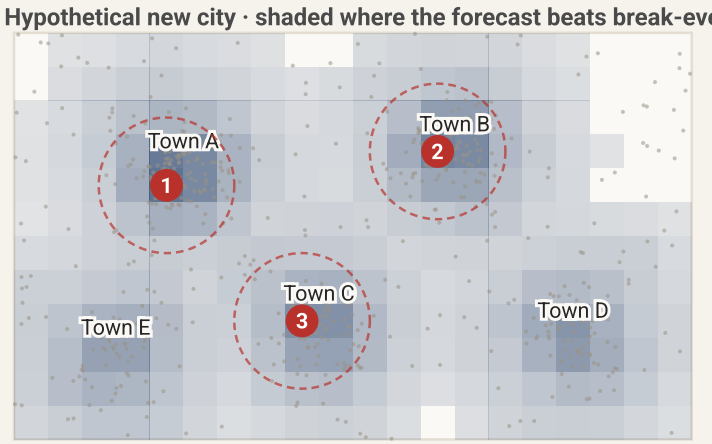
Trained on Singapore's outlets, the model picks where to open in a city where Chagee has no data yet

Supervised learning: keep only the inputs a new city has, test the model on outlets it has not seen, then let it score every spot in a hypothetical city (illustrative, synthetic).

TRAIN ON 44 OUTLETS · TOGGLE THE INPUTS

- ☑ **People within reach** — NEW MARKET: HAS IT
- ☑ **Distance to a town centre** — NEW MARKET: HAS IT
- ☐ Delivery-area demand — NEW MARKET: TO CONFIRM
- ☐ Unserved app searches — NEW MARKET: NONE
- ☐ Chagee outlets nearby — NEW MARKET: NONE
- ☐ Distance to nearest Chagee outlet — NEW MARKET: NONE
- ⬚ Competitors within 500 m — NOT IN OUR DATA
- ⬚ Rent per sq ft — NOT IN OUR DATA





TESTED ON OUTLETS IT HAS NOT SEEN

±83 cups a day, average miss

R^2 0.76 with 2 inputs

All inputs (Singapore)	±41 · R^2 0.95
New-market inputs only	±83 · R^2 0.76
No inputs: guess the average outlet	±197

WHERE TO OPEN IN THE NEW CITY

1. Town A: ≈ **1,090** ± 83 · **above break-even even at 1,007** · **beyond what it learned**
2. Town B: ≈ **959** ± 83 · **above break-even even at 876**
3. Town C: ≈ **887** ± 83 · **above break-even even at 804**

Cups a day, forecast ± average miss. Break-even ≈ 275 at Singapore costs (slide 09), a stand-in. Picks ≥ 2 km apart.

WHAT IS AI HERE Supervised learning (ridge regression) from site attributes to cups a day. The labels are the slide 05 engine's synthetic outlets, so the fit flatters; Chagee's real opening history would fit worse. Using Singapore to score another market assumes its customers behave alike: check against the first outlets' actuals.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

SPEAKER NOTES · THE SUPERVISED MODEL

This is the model behind step three of the framework, actually running. It learns cups a day from a site's attributes, trained on Singapore's 44 outlets. The labels are synthetic, from the slide 05 engine, so read it as the method, not a forecast.

We test it fairly: each outlet is predicted by a model that never saw it. With every input, it misses by about 41 cups a day on average. Now the proxy for a new market: there is no Chagee app history and no outlet network yet, so switch those inputs off. With only population and distance to a town centre, the error grows to about 83 cups a day, still better than guessing the average outlet, about 197.

Then the point of it: where to open in a place with no local data. The right-hand map is a hypothetical city, generated at the same density as our Singapore data. With only the inputs that city has, the model scores every spot and picks the best sites at least 2 km apart, so they do not take sales from each other. Each pick shows its forecast with the error band, and whether it clears break-even at Singapore costs, which stand in until local costs are known. Picks beyond what the model saw in Singapore are flagged. So the model can travel, but only on inputs the new market has, with wide ranges: pilot the top site first and recheck the model against its actuals. That is the continual-learning step on the next slide. About fifty seconds.

Extension: where this scales next. Not included in the S$ benefit.

From pin to net profit

Six steps from a pin to a go / no-go, on profit net to Chagee overall (illustrative). Decision-maker here: corporate expansion planning, not the GM.

1 Pin a location
Corporate clicks a point on the map.
NOT AI

2 Tag attributes
People, transit, footfall, delivery, competitors, outlets nearby, rent.
DATA

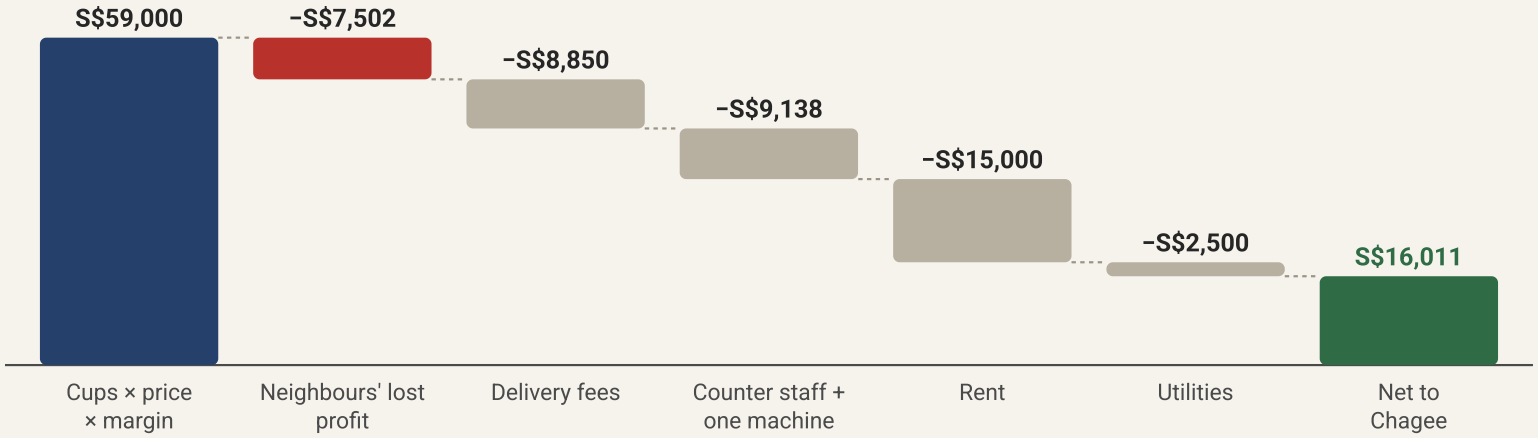
3 Four scores
Demand, delivery, competition, cannibalisation (slide 08).
AI · SUPERVISED MODEL

4 Find look-alikes
Anonymised outlets (Outlet 12, Outlet 27): their cups a day, what neighbours lost.
AI · SIMILARITY SEARCH

5 Forecast net profit
Cups a day, running cost, minus the profit neighbours lose.
COMPUTED BELOW

6 Decide
Open, pilot or skip. Actuals feed back: continual learning.
AI · CONTINUAL LEARNING

NET PROFIT TO CHAGEE, A MONTH · SITE A, TOP APP-SEARCH CLUSTER



At 278–833 cups a day: **−S$3,506** to **S$34,624** a month. Neighbour outlet B, 2.7 km away, loses 9% of its sales. Delivery fees: 30% of cups × 30% commission. Counter: the one-machine setup, normal-day roster (counter model).

STEP 6 · RECOMMENDATION

Pilot

positive at 556 cups a day, negative at 278, the low end: test it before a full lease.

Site score **76 / 100 (slide 08)**
Payback **≈ 25 months on S$400,000**

A strong-demand site can still fail if rent or staffing is too high, or if it mostly takes sales from our own neighbours.

In practice this needs Chagee's opening history. With about 44 outlets in Singapore the model must stay simple (few attributes, ranges not point values).

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it. Cups a day, price and margin are the benefit slide's (slide 11); rent, delivery, utilities and set-up cost are assumptions, not Chagee data.

SPEAKER NOTES · FROM PIN TO NET PROFIT

Six steps from a pin to a go or no-go. Steps three, four and six are where the AI would sit in production: a supervised model for the scores, a similarity search for look-alike outlets, and continual learning as each new outlet reports actuals. On this slide only step five is computed.

The waterfall starts from cups a day times price times margin, the same inputs as the benefit slide: 556 cups a day. Take off the profit neighbours lose, delivery fees, the counter, rent and utilities: about S$16,011 a month net to Chagee, a payback of about 25 months. At the low end of the cups range it loses money, so the call is: Pilot. Move rent, delivery or set-up cost and watch the call change; these are our assumptions, not Chagee data.

The honest point: switch to the pin with an outlet 1.7 km away. It takes 44% of its cups from our own outlets and comes out at −S$2,697 a month, a skip. About forty-five seconds.

Later: reuse in new markets via transfer learning, out of scope.

 TBC · DRAFT, TO BE CONFIRMED

Four decisions for the GM, each from a simulation, then a pilot to test them

	DECISION	WHY, FROM THE MODEL	REVISIT WHEN
1 **Where to open** slide 05 · DBSCAN + assumed demand model	**Open where app searches cluster, not beside an existing outlet.**	**+731** against **+216** cups a day for the same new outlet. Synthetic app-search data: this shows the method, not a finding about real outlets.	Rerun on Chagee's app logs, if searches and location are recorded, before any lease.
2 **What to automate** slide 07 · physical AI	**In our model, one tea-and-milk machine is the cheapest on every day type; pilot it against today's split (our scenario).**	**S$305–335** a day against S$372–402. The two cost the same at about **S$905 a month per machine unit** (1 unit against 3), with wage, volume, station speeds and crews held fixed; we assume S$1,909. Amacha's S$10,000 is a one-off price; without machine life, upkeep and financing we cannot turn it into a monthly cost.	When Chagee's real machine cost is known.
3 **How many people** slides 06, 11 · forecast + economic rule	**Economic rule: add a person on event days, not on busy days.**	With the one machine: peak crew 2 → **3** on event days; 2 on busy days, where an extra person costs more than the orders it keeps.	When the forecast peak moves.
4 **How to share** slide 06 · operations	**Test a bucket brigade against fixed posts for the hand steps.**	In the recommended setup (one machine, 3 people, event day): **0.2%** walk away with a brigade, 0.4% with well-assigned fixed posts, 22% with badly assigned ones; median wait 1:44 against 2:18.	Fixed posts once there is spare staff.

- **Pilot in 3 outlets**

  Shadow forecast only | **Roster** economic rule | **Machine** one vs today's split

  KPIs: cups per labour hour, wait time and walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

  All figures are illustrative assumptions, not Chagee data. Machine costs and speeds are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

  The real machine cost per unit, whether outlets log wait times and walk-aways, and how today's local staffing calls compare with the forecast.

ASSUMED BASELINE illustrative — One machine and a flat crew of 2 every day: **23%** of event-day peak orders walk away.

PROPOSED SCENARIO in our model — The forecast and the economic rule add a person on event days only (2 → 3): **0.2%** walk away; busy days stay at 2 (0.1%).

SPEAKER NOTES · THE GM'S PLAYBOOK

TBC: this slide is a draft and still to be confirmed by the team; present it as work in progress.

Four decisions, one per simulation, all in our model and on our assumptions. One, where to open: DBSCAN clusters app searches and our assumed demand model scores the sites. The searches are synthetic, so this shows the method; rerun it on Chagee's app logs, if searches and location are recorded, before any lease.

Two, what to automate: in our model one tea-and-milk machine is the cheapest on every day type, so pilot it against today's split, which is our scenario, not Chagee's disclosed setup. The two cost the same at about S$905 a month per machine unit, comparing one unit against three, with the wage, volume and crews held fixed. We assume S$1,909. We cannot turn the one-off price one small shop paid into a monthly cost without machine life, upkeep and financing, so Chagee's real cost decides this.

Three, how many people: the economic rule adds a person on event days but not on busy days. Four, how to share: test a bucket brigade against fixed posts; in the recommended setup the brigade does a little better than well-assigned fixed posts and much better than badly assigned ones. Then a pilot in 3 outlets: shadow the forecast, roster by the economic rule, and run one machine against today's split, judged on cups per labour hour, wait time and walk-aways. About one minute.

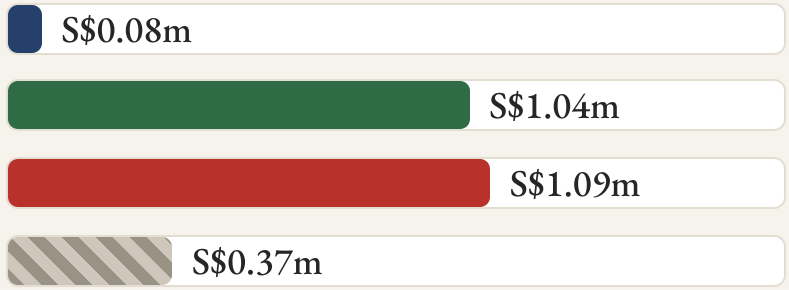
 TBC · DRAFT, TO BE CONFIRMED

In our model, about S$1.09m a year across 44 outlets against an illustrative baseline, almost all from one machine's labour saving

Modelled avoided costs on stated assumptions, not demonstrated savings and not savings against Chagee's actual operations, where the GM and teams staff on local expertise. The pilot compares one machine with today's split (our scenario).

PROFIT A YEAR, ALL 44 OUTLETS · ECONOMIC RULE THROUGHOUT

AGAINST THE ILLUSTRATIVE ALL-HAND BASELINE (FLAT NORMAL-DAY ROSTER)



PILOT COMPARISON: THE PROPOSED SCENARIO AGAINST TODAY'S SPLIT (OUR SCENARIO), BOTH ON THE ECONOMIC RULE



Profit = orders kept × price × margin, plus labour saved, minus machine cost. Calendar: 104 busy days (weekends) and 11 event days (public holidays) a year, an illustrative definition. Baseline: all by hand, the normal-day crew every day. Peak hour only, so orders kept are a floor. Every figure, range and both crew rules: appendix.

PHYSICAL AI · ONE TEA-AND-MILK MACHINE

Most of the benefit: labour

Lets the roster drop from 2 people to 1 in the off-peak hours: **S$1.04m** a year after the machine's cost.

FORECAST + ECONOMIC RULE

Adds event-day orders on top of the machine

Adds a person on event days only (2 → 3), keeping about 35 more orders per outlet that day: 390 a year, **S$0.05m** across Singapore. On its own, all by hand: S$0.08m.

READ WITH CARE

The big number rests on one assumption

The minimum crew, 2 by hand and 1 with machines, is ours. Without it, S$1.09m falls to **S$0.37m**. The pilot arm beats today's split by S$1.22m: more than the headline, because today's split is itself S$0.14m below the baseline.

ASSUMED BASELINE
illustrative

Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

PROPOSED SCENARIO
in our model

Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE SIZE OF THE BENEFIT

TBC: this slide is a draft and still to be confirmed by the team; present it as work in progress.

One number and where it comes from, all in our model. Against an illustrative baseline, an all-by-hand counter with the normal-day crew every day, the proposed scenario gives about S$1.09m a year across 44 outlets. That baseline is ours, not Chagee's operations, so these are not savings Chagee would see; they are modelled avoided costs on stated assumptions. Every figure uses the economic rule, the rule the playbook recommends.

Read the bars. Almost all of it is the one tea-and-milk machine, which lets the roster drop a person off-peak. Rostering to the forecast by the economic rule adds a person on event days only and keeps about 35 more orders per outlet that day, S$0.05m a year across Singapore. The last bar is the pilot comparison: the proposed scenario against today's split, our scenario, both on the economic rule, a difference of S$1.22m.

Say the caveat out loud: the machine's saving comes from our minimum-crew assumption; without it, the total falls to about S$0.37m. Ranges and both crew rules are in the appendix. About one minute.

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



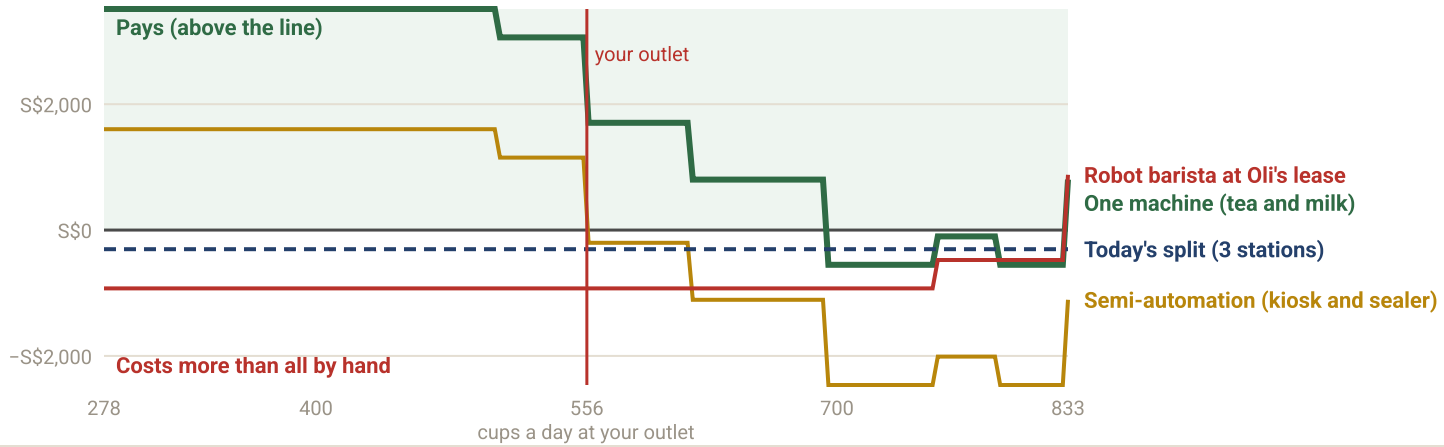
SPEAKER NOTES · THANK YOU

Likely questions: is the brigade always better? No; with spare staff fixed stations are as good. Is the forecast real? No, it is an illustration of the chain, built on stated assumptions.

A machine pays only when it removes a person-shift for less than the shift costs

To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET



Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

	1 Semi-automation	2 Beverage platform	3 Milk-tea machine	4 Robot arm	5 Robot barista
	Gong Cha Singapore	Gong cha Super Wu	Amacha, one shop	Richtech ADAM	Oli; coffee-first
	Not disclosed	*Not disclosed*	S$10,000 to buy	*Not disclosed, rental only*	US$4,990/mo lease
	PRESS	COMPANY CLAIM	PRESS	VENDOR PAGE	VENDOR PAGE

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

One machine on the tea and milk

+S$1,706 a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

Today's 3-station split

−S$305 a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

Robots: arm or barista

757+ cups a day before they save more labour than today's split (labour only, not a cost break-even)

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

Model assumption that drives all three: minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

WITHOUT AI
our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

WITH AI
our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. A robot setup (four stations on machines) starts to save more labour than today's split at about 757 cups a day; labour only, not a cost break-even.

SPEAKER NOTES · MACHINE LADDER

Backup for slide 07, for questions on which machine to buy. This is physical AI: automation and machines at the counter. The predictive AI is the forecast that sets the roster. One rule explains the whole slide: a machine pays only when it removes a person-shift, and costs less than that shift.

The chart shows the net gain a month against an all-by-hand counter as volume changes; above the line it pays. One tea-and-milk machine lets the roster drop from 2 people to 1 in 8 off-peak hours, about one person-shift a day, for one station unit: it pays across most of the range. Today's split saves 12 person-hours but pays for 3 units, so it ends up a little dearer than all by hand. A robot setup, four stations on machines, saves no more labour than today's split until about 757 cups a day. That is where extra labour saving begins, not a cost break-even; at Oli's lease it only pays at the top of our range.

Three things to say out loud. The minimum crew, 2 by hand and 1 with machines, is our model's assumption and drives this result. Every option runs at the model's station speeds, not vendor claims. Payback is simple payback, if the machine lets the roster drop one person-shift, with no installation, maintenance or financing. Chagee's own equipment cost is not public. About one minute.

Takeaway: At 556 cups a day: start with one machine (about +S$1,706 a month); today's 3-station split costs S$305 a month more than all by hand; robots save no more labour than today's split until about 757 cups a day.

Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

Modelled avoided costs on stated assumptions, not demonstrated savings, against an illustrative baseline, not Chagee's operations. Economic rule for the headline figures. Peak hour only; conservative. Prices to our knowledge, based on public sources.

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

EVENT DAY
61 orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

BUSY DAY
5.4 orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

NORMAL DAY
0.2 orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

A YEAR	PER OUTLET	44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL
Orders kept peak hour, busy and event days	**1,278** range −121 to 1,452	**56,221**
Revenue orders × price	**S$7,539** range −S$832 to S$10,015	**S$331,704**
Profit margin 50%–70%: −S$754 to +S$754	S$24,660	S$1,085,027
Labour saving, counter labour saved − machine cost; shown on its own	S$20,136	S$886,005

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

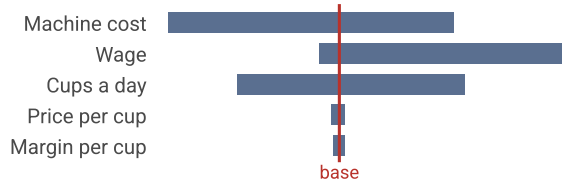
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

ORDERS · PROFIT A YEAR	WALK-AWAY RULE	ECONOMIC RULE
Forecast crew	1,242 · S$603	670 · S$1,711
Machine	888 · S$23,611	888 · S$23,611
Both, one run	**1,278 · S$24,660**	**1,278 · S$24,660**
Both, today's split counter −S$305/month	1,124 · −S$3,142	1,124 · −S$3,142

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (−S$662 profit against the sum). Headline figures use the economic rule.

Against what
A flat normal-day roster and an all-by-hand counter. Chagee deploys people on local expertise (Chagee source, anonymous), so this baseline is ours, not Chagee's.

All inputs at their worst/best end together
Profit a year, one outlet: low **−S$15,660** · base **S$24,660** · high **S$101,166**



Fragile: the minimum crew
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

Not sized
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

ASSUMED BASELINE illustrative
Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

PROPOSED SCENARIO in our model
Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

SPEAKER NOTES · THE BENEFIT IN DETAIL

Backup for slide 11. This answers part c of the brief: how big is the benefit, for Singapore only. Against what: a flat normal-day roster and an all-by-hand counter. Chagee deploys people on local expertise, the GM and outlet teams making the calls, so this baseline is ours, not Chagee's practice.

Lead with orders. On an event day the model keeps about 61 orders per outlet in the peak hour that would otherwise walk away, about S$358 of revenue; on a busy day about 5. Over a year it depends on how many such days there are: our illustrative calendar is 104 busy days, the weekends, and 11 event days, the public holidays. That is an input, not a measurement.

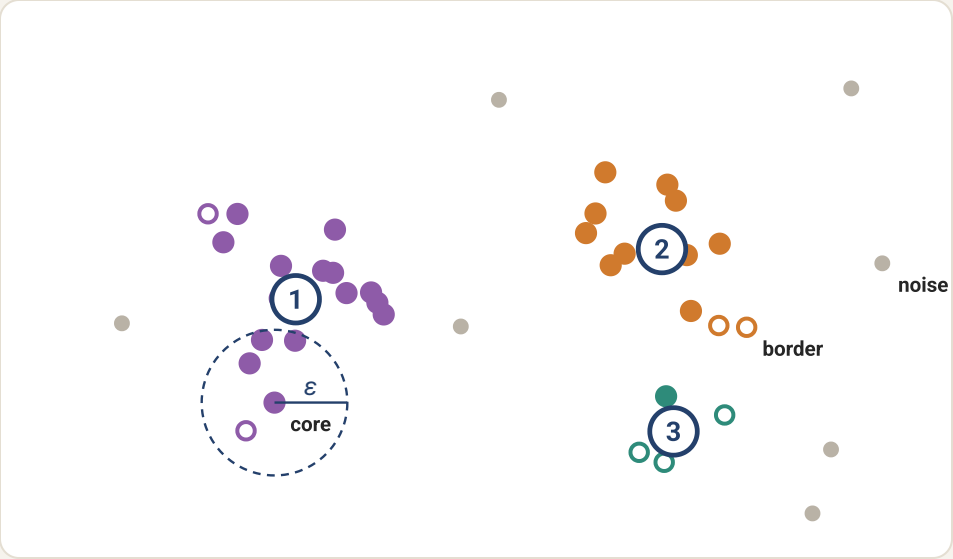
Profit is the second line, using a margin we assume because Chagee does not disclose one. The labour saving is shown separately and never added to revenue. The two levers are run together, not added, because they share the same labour hours. The strip at the top splits the profit: most is counter automation, physical AI, the machine on its own; the forecast, predictive AI, adds a small share on top.

Two crew rules for rostering to the forecast. The walk-away rule adds people until peak-hour walk-aways are within 2%; at our default volume that is exactly the forecast crew. The economic rule adds a person only when the orders kept are worth more than the extra hours. They differ on busy days by hand, where the economic rule adds no one.

Be honest about fragility: most of the profit is the counter's labour saving, and it comes from the model's minimum crew, 2 by hand and 1 with machines. Drop that rule and the profit falls to about S$8,395 a year per outlet. And rostering to the forecast pays on event days, not on busy days, where the extra hours cost more than the orders won back. These are modelled avoided costs on stated assumptions, not demonstrated savings. About one minute.

DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 05 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

SPEAKER NOTES · HOW THE SITE FORECAST WORKS

Backup for questions on slide 05. Left: DBSCAN in four moves on a toy set. A search is a core point if enough searches sit within the radius; clusters grow from core to core; border points join; the rest is noise. We do not choose the number of clusters.

Right: every simulated number, in the order the model uses it. The forecast is only as good as these assumptions; with Chagee's real app logs, the first two rows would be replaced by data and the rest would run as is.

Every rung, every price as published, and where it comes from

To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.

RUNG	EXAMPLE	AUTOMATES · STEPS ON MACHINES (OUR MAPPING)	VENDOR LIST PRICE	CLAIMS (NOT MODEL INPUTS)	RELIABILITY	SOURCE
1 **Semi-automation**	Gong Cha Singapore	Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled.	*Not disclosed*	Over 20% productivity from self-ordering; 5% manpower saving (no baseline given)	PRESS	https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/
2 **Beverage platform**	Gong cha Super Wu (Feb 2026)	Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk**	*Not disclosed*	Up to 65% higher peak productivity	COMPANY CLAIM	https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/
3 **Single automated milk-tea machine**	Amacha (one small shop, owner interview)	Makes the milk tea itself. **Tea and milk**	S$10,000 to buy	The shop's total opening cost was S$209,000 (not a machine cost)	PRESS	https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715
–	**Our scenario: today's split**	**Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 06 and 07), not Chagee's disclosed setup	Model: S$1,909 a month per station unit		OUR ASSUMPTION	assumptions.json
4 **Robot arm that makes boba**	Richtech ADAM	Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal**	*Not disclosed*. Rental only; price not published	–	VENDOR PAGE	https://richtechrobotics.com/solutions/adam
5 **Robot barista**	Oli	A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly.	US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Artly		US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Cafe X		US$250,000 (≈ S$318,125) to buy Oli's estimate of a competitor's price.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
–	**Chagee Singapore**	Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment.	*Not public*	–	PRESS	https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/

Vendor list prices, not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

SPEAKER NOTES · MACHINE LADDER: TIERS AND SOURCES

Backup for the machine ladder. Every price is as the source publishes it; where a source gives no price we say not disclosed. Claims such as Gong cha's productivity figures are listed for context and are not inputs to our model. Soft copy only.

Where every number comes from

FACTS QUOTED

Outlets in Singapore: 44. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

Franchise share of FY2025 net revenue: 0.88. Secondary source; flagged wherever shown. *[L-koalagains]*

Company-owned share of FY2025 net revenue: 0.12. Secondary source; flagged wherever shown. *[L-koalagains]*

Registered members, group-wide, end of June 2026: 257000000. Company statement on the Q2 2026 earnings call (COO, interpreted). Group-wide, not Singapore. Membership is on Chagee's mobile mini program (L-chagee-q1-2025). *[L-chagee-q2-2026-call]*

Orders from members with two or more purchases, group-wide (more than): 0.78. Company statement on the Q2 2026 earnings call. A floor ('more than 78%'). Group-wide, not Singapore. *[L-chagee-q2-2026-call]*

SOURCES

L-sg-directory Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-koalagains KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-yfchuhai YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-sec20f Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-unsw UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

I-chagee-contact CHAGEE SOURCE, ANONYMOUS (Oct 2026): a Chagee source who asked not to be named; do not identify them. Four points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work, because the data sits in one system; Singapore outlets run under one company and one data system; all delivery is via third-party apps; people are deployed on local expertise: the GM and outlet teams make the staffing calls.

I-sg-groundfloor INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

I-team-app-first INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing. SUPERSEDED on slide 01 (22 Oct 2026) by P-appstore, P-vulcanpost-comeback and L-sec20f-data; the app's share of orders stays to confirm.

L-ester1996 Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 05.

P-honeycombers PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

P-eatbook PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

I-team-expansion INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

I-team-site-framework TEAM WORK (Oct 2026): Jin's site-selection framework, two team slides ('Four Scores for Every Location', 'From Pin to Profit Forecast'): four scores per pin, six steps from a pin to a go / no-go on profit net to Chagee. Used for slides 08 and 09 (extension). Its example values and outlet names are not used; ours are illustrative.

L-chagee-q2-2026-call Chagee (CHA) Q2 2026 Earnings Call, full transcript, TradingKey, 29 Aug 2026 (local copy: 'Chagee (CHA) Q2 2026 Earnings Call_ Profit Margin Improves as Overseas GMV Doubles.pdf', supplied by the team). COO's remarks, interpreted: 'As of the end of June, our total registered members reached 257 million. The repurchase rate among active members remain above 43%, while members who made 2 or more purchases accounted for more than 78% of total orders.' Group-wide, not Singapore. The page's AI summary is not used; the quote is from the transcript.

L-chagee-q1-2025 Chagee Holdings, 'Chagee Announces First Quarter 2025 Unaudited Financial Results', GlobeNewswire, 30 May 2025 (local copy supplied by the team; the team's link https://investor.chagee.com/node/6736/pdf blocks automated access, so not checked by us). 'Chagee's Mobile Mini Program had a total of 192.4 million registered members as of March 31, 2025'; active members = registered members who placed an order at least once in a period. Shows that membership sits on Chagee's ordering app. Group-wide.

P-appstore CHAGEE app, App Store listing (team research, Oct 2026; no link supplied, not checked by us): 'Order and pay digitally', 'Order for self-pickup ahead of time', 'Locate our outlets and their operating hours'. The listing seen was the Saudi storefront; its privacy-policy URL includes 'SGMY' (our reading: Singapore and Malaysia, an inference). Privacy label: Precise Location under 'Data Not Linked to You'; search history not listed.

P-vulcanpost-comeback PRESS - Vulcan Post, Jayme Teo, 2 Oct 2024, https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/ : after the reopening, with some customers 'waiting for as long as an hour to receive their orders', 'the brand rolled out in-app ordering and delivery and constantly upgraded the ordering process.' Its 'kiosks' are a store format ('pivoting from their takeaway kiosks for comfortable dine-in, sit-down tea bars'), not ordering kiosks. Quotes checked by us, 22 Oct 2026.

L-sec20f-data Chagee Holdings Form 20-F for FY2025 (same filing as L-sec20f), risk factors: 'Our business generates and processes personal, transactional and other data.' and 'we may not be able to use our membership program to deepen our understanding of our target consumers.' No share of orders by app or mini program, and nothing on search or location data, in the parts we read. Quotes checked by us, 22 Oct 2026.

ASSUMPTIONS (ASSUMPTIONS.JSON, ILLUSTRATIVE)

Cups per outlet per year: 200000 cups/store/year

Share of the day's cups in each open hour: 0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07 share of daily cups per open hour (10:00 to 22:00), sums to 1

Busy and event day multipliers: none 1, moderate 1.5, large 2.5 x forecast arrival rate

Busiest hours the multiplier lands on: 2 busiest forecast hours that get the surge

Seconds per cup by hand, per station: ticket 5, cup ice 10, tea milk 15, topping finish 12, shake seal 10, handover 8 seconds of one average worker per cup

Seconds per cup by robot, per station: ticket 2, cup ice 8, tea milk 10, topping finish 15, shake seal 8, handover 6 seconds per cup per machine unit

Stations the robots do today: ticket, tea milk, shake seal

Spread of worker speeds: 0.2 +/- share around average speed

Minute-to-minute variation in pace: 0.25 coefficient of variation, minute to minute

Share of new staff: 0.25 share of roster in first months

New staff speed: 0.7 x average speed

Coordination loss, people around robots: 0.9 multiplier on human-pool throughput in the mixed setup

Minimum people on the counter: all human 2, mixed 1, all machine 0 workers per open hour

Roster safety margin over the forecast: 1.2 x forecast peak rate

Labour cost per hour worked (SGD): 15.06 SGD per hour worked, fully loaded (statutory costs)

SPEAKER NOTES · SOURCES AND ASSUMPTIONS

Soft copy only.